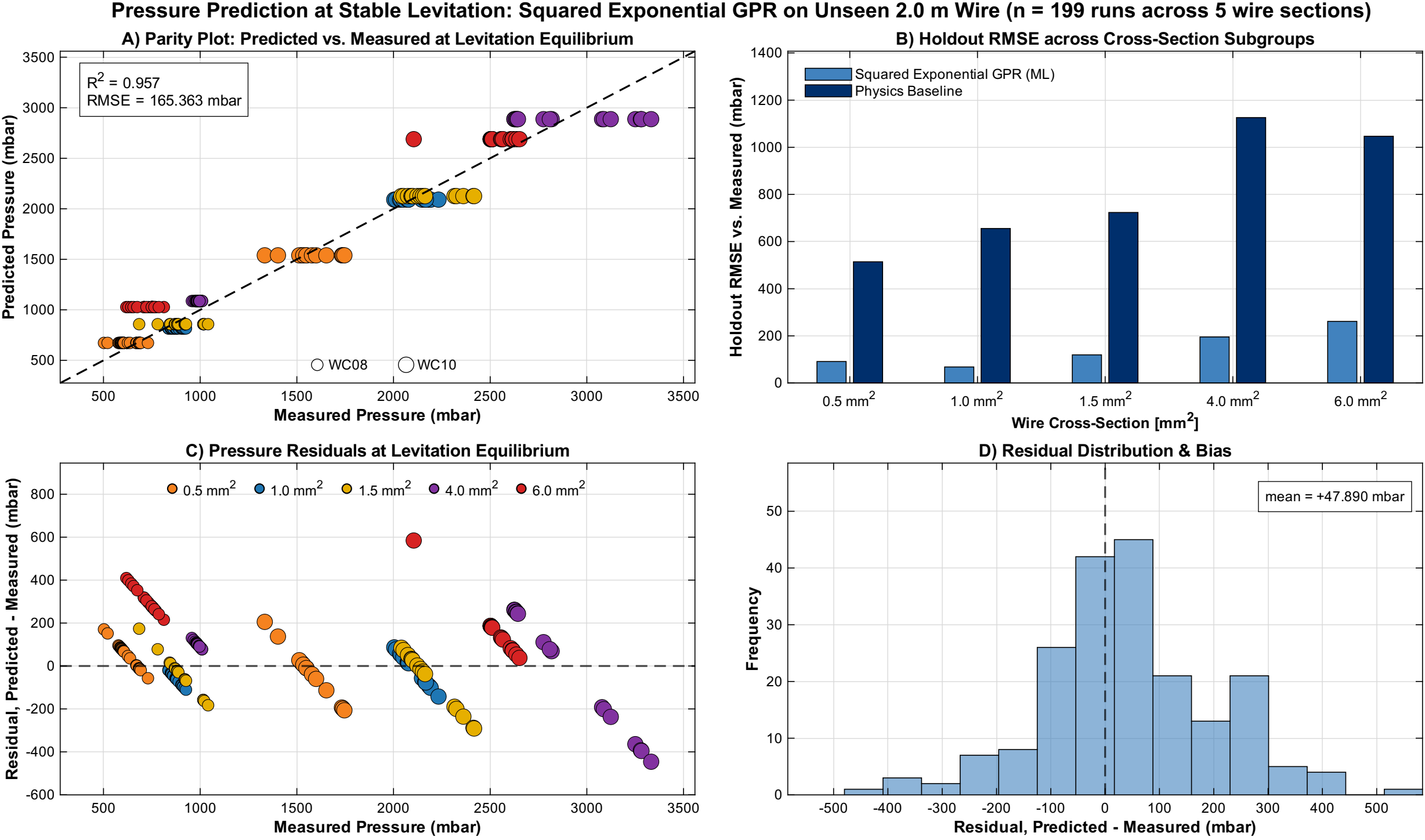

Pressure Prediction at Stable Levitation: Squared Exponential GPR on Unseen 2.0 m Wire (n = 199 runs across 5 wire sections)
A) Parity Plot: Predicted vs. Measured at Levitation Equilibrium
R2 = 0.957
RMSE = 165.363 mbar
WC08
WC10
Predicted Pressure (mbar)
Measured Pressure (mbar)
B) Holdout RMSE across Cross-Section Subgroups
Squared Exponential GPR (ML)
Physics Baseline
Holdout RMSE vs. Measured (mbar)
0.5 mm2
1.0 mm2
1.5 mm2
4.0 mm2
6.0 mm2
Wire Cross-Section [mm2]
C) Pressure Residuals at Levitation Equilibrium
Residual, Predicted - Measured (mbar)
Measured Pressure (mbar)
D) Residual Distribution & Bias
mean = +47.890 mbar
Frequency
Residual, Predicted - Measured (mbar)